

Organic synthesis

A Level Chemistry

Organic synthesis

Like the AS synthesis topic, this asks you to **join up** known reactions to build a target molecule. Now you also have the A Level reactions of arenes, phenol, amines, amides and acyl chlorides.

Identifying functional groups

A molecule may carry several **functional group** 官能团 types. Use the test reactions to spot each one, then predict its behaviour. For example:

- decolourises bromine water with **no** catalyst, giving a white precipitate → a phenol or a phenylamine (the ring is activated).
- reacts violently with cold water, giving fumes of HCl → an **acyl chloride** 酰氯.
- gives a purple colour with neutral FeCl_3 → a **phenol** 苯酚.

A map of the A Level reactions

Start	Reagent and conditions	Product
benzene (an arene 芳烃)	$\text{HNO}_3 / \text{H}_2\text{SO}_4$ (electrophilic substitution 亲电取代)	nitrobenzene
nitrobenzene	$\text{Sn} / \text{conc HCl}$, then NaOH (reduction 还原)	phenylamine
phenylamine	HNO_2 , below 10°C	a diazonium salt
diazonium salt	warm water	phenol; or couple with phenol → azo dye
methylbenzene	hot KMnO_4 (oxidation 氧化)	benzoic acid
carboxylic acid 羧酸	SOCl_2 or PCl_5	acyl chloride
acyl chloride	alcohol / phenol	an ester 酯
acyl chloride	ammonia / amine 胺	an amide 酰胺
amide or nitrile 腈	LiAlH_4	an amine
halogenoalkane	KCN	a nitrile (adds one carbon)

Planning and analysing a route

To devise a **synthetic route** 合成路线, work **backwards** from the target: which single reaction makes it, and from what? Repeat until you reach the starting material, then write each step with its **reagent** 试剂 and conditions.

When you **analyse** a route, state the type of reaction for each step (for example electrophilic substitution, **addition-elimination** 加成消去, oxidation or reduction) and watch for likely **by-products** 副产物—for example, making an amine from a halogenoalkane also gives over-substituted amines, lowering the yield.